JEE–MAIN EXAMINATION – JANUARY 2026

(HELD ON SATURDAY 24th JANUARY 2026) **TIME : 3:00 PM TO 6:00 PM**

CHEMISTRY

TEST PAPER WITH ANSWER

SECTION-A

51. Choose the **INCORRECT** statement

(1) Among the isotopes of carbon, ^{13}C is a radioactive isotope.

(2) Carbon exhibits negative oxidation states along with +4 and +2.

(3) Carbon cannot exceed its covalency more than four.

(4) CO_2 is the most acidic oxide among the dioxides of group of 14 elements.

52. The unsaturated ether on acidic hydrolysis produces carbonyl compounds as shown below:-

$$CH_3 - CH = CH - O - CH = CH_2$$

$$\downarrow H_3O^{(+)}$$

$$CH_3 - CH_2 - CH = O \text{ and } O = CH - CH_3$$

Based on this, predict the solution / reagent that will help to distinguish "P" and "Q" obtained in the following reaction.

$$CH_3 - CH = CH - O - \underset{\displaystyle CH_3}{\overset{\displaystyle |}{C}} = CH_2 \xrightarrow[\Delta]{H_3O^{(+)}} P + Q$$

(1) Lucas reagent

(2) 2,4-DNP reagent

(3) Saturated $NaHSO_3$ solution

(4) Fehling solution

53. The number of possible tripeptides formed involving alanine (ala), glycine (gly) and valine (val), where no amino acid has been used more than once is :

(1) 6 (2) 3

(3) 4 (4) 8

54. Two liquids A and B form an ideal solution at temperature T K. At T K , the vapour pressures of pure A and B are 55 and 15 kNm^{-2} respectively. What is the mole fraction of A in solution of A and B in equilibrium with a vapour in which the mole fraction of A is 0.8 ?

(1) 0.5217 (2) 0.480

(3) 0.663 (4) 0.340

55. Consider the following gaseous equilibrium in a closed container of volume "V" at T(K).

$$P_2(g) + Q_2(g) \rightleftharpoons 2PQ(g)$$

2 moles each of $P_2(g)$, $Q_2(g)$ and PQ (g) are present at equilibrium. Now one mole each of 'P_2' and 'Q_2' are added to the equilibrium keeping the temperature at T(K). The number of moles of P_2, Q_2 and PQ at the new equilibrium, respectively , are -

(1) 2.67, 2.67, 2.67 (2) 1.21, 2.24, 1.56

(3) 1.66, 1.66, 1.66 (4) 2.56, 1.62, 2.24

56. Pair of species among the following having same bond order as well as paramagnetic character will be-

(1) O_2^+, N_2^{2-} (2) O_2^-, N_2^+

(3) O_2^+, N_2^- (4) O_2^-, N_2^-

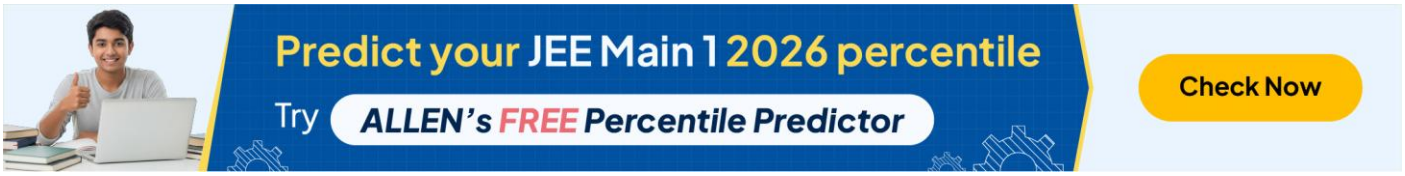
57. The wavelength of light absorbed for the following complexes are in the order.

$$\underset{(I)}{[Co(NH_3)_6]^{3+}} ; \underset{(II)}{[Co(H_2O)_6]^{3+}} ; \underset{(III)}{[Co(CN)_6]^{3-}} ;$$

$$\underset{(IV)}{[Co(NH_3)_5(H_2O)]^{3+}} ; \underset{(V)}{[CoF_6]^{3-}}$$

(1) III < I < II < IV < V (2) III < I < IV < V < II

(3) III < IV < I < II < V (4) III < I < IV < II < V



58. One mole of Cl_2 (g) was passed into 2 L of cold 2M KOH solution. After the reaction, the concentrations of Cl^-, ClO^- and OH^- are respectively (assume volume remains constant)

(1) 0.75 M , 0.75 M , 1 M

(2) 0.5 M , 0.5 M , 0.5 M

(3) 0.5 M , 0.5 M , 1 M

(4) 1 M , 1M , 1 M

59. A student has planned to prepare acetanilide from aniline using acetic anhydride. The student has started from 9.3g of aniline. However, the student has managed to obtain 11 g of dry acetanilide.

The % yield of this reaction is :-

(1) 81.5% (2) 97.5%

(3) 59.5% (4) 72.5%

60. Find out the statements which are **not** true.

A. Resonating structure with more number of covalent bonds and lesser charge separation are more stable.

B. In electromeric effect, an unsaturated system shows + E effect with nucleophile and –E effect with electrophile.

C. Inductive effect is responsible for high melting point, boiling point and dipole moment of polar compounds.

D. The greater the number of alkyl groups attached to the doubly bonded carbon atoms, higher is the heat of hydrogenation.

E. Stability of carbanion increases with the increase in s-character of the carbon carrying the negative charge.

Choose the ***correct*** answer from the options given below.

(1) A, D & E only (2) B, D & E only

(3) A, C & D only (4) B & D only

61. The correct order of C, N, O and F in terms of second ionisation potential is

(1) F < N < C < O

(2) C < O < N < F

(3) C < N < F < O

(4) C < F < N < O

62. In the Group analysis of cations , Ba^{2+} & Ca^{2+} are precipitated respectively as

(1) sulphide & sulphide

(2) hydroxide & carbonate

(3) carbonate & carbonate

(4) chromate & sulphide

63. The wavelength of spectral line obtained in the spectrum of LI^{2+} ion, when the transition takes place between two levels whose sum is 4 and difference is 2, is

(1) 2.28×10^{-7} cm

(2) 2.28×10^{-6} cm

(3) 1.14×10^{-7} cm

(4) 1.14×10^{-6} cm

64. Given below are two statements :

Statement I : Cross aldol condensation between two different aldehydes will always produce four different products.

Statement II : When semicarbazide reacts with a mixture of benzaldehyde and acetophenone under optimum pH, it forms a condensation product with acetophenone only.

In the light of the above statements, choose the ***correct*** answer from the options given below :

(1) Both Statement I and Statement II are false

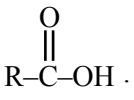
(2) Statement I is false but Statement II is true

(3) Both Statement I and Statement II are true

(4) Statement I is true but Statement II is false



65. Given below are two statements :

Statement I : The dipole moment of R-CN is greater than R-NC and R-NC can undergo hydrolysis under acidic medium to produce  .

Statement II : R-CN hydrolyses under acidic medium to produce a compound which on treatment with $SOCl_2$, followed by the addition of NH_3 gives another compound(x). This compound (x) on treatment with NaOCl/NaOH gives a product, that on treatment with $CHCl_3$/KOH/Δ produces R-NC

In the light of the above statements, choose the ***correct*** answer from the options given below :

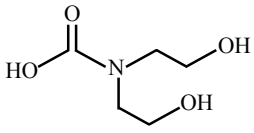
(1) Both Statement I and Statement II are false

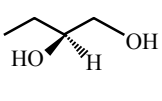
(2) Both Statement I and Statement II are true

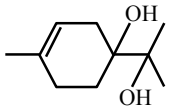
(3) Statement I is true but Statement II is false

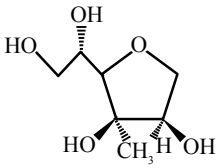
(4) Statement I is false but Statement II is true

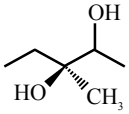
66. From the following, how many compounds contain at least one secondary alcohol ?

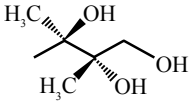

(I)


(II)


(III)


(IV)


(V)


(VI)

Choose the ***correct*** answer from the options given below :

(1) Five (2) Three

(3) Four (4) two

67. The heat of atomisation of methane and ethane are 'x' kJ mol^{-1} and 'y' kJ mol^{-1} respectively. The longest wavelength (λ) of light capable of breaking the C-C bond can be expressed in SI unit as :

(1) $\frac{hc}{1000}\left(\frac{y-6x}{4}\right)^{-1}$

(2) $\frac{N_A hc}{250(4y-6x)}$

(3) $\frac{N_A hc}{250(y-6x)}$

(4) $N_A hc\left(y-\frac{6x}{4}\right)^{-1}$

68. At 298 K, the mole percentage of $N_2(g)$ in air is 80%. Water is in equilibrium with air at a pressure of 10 atm. What is the mole fraction of $N_2(g)$ in water at 298 K ? (K_H for N_2 is 6.5×10^7 mm Hg)

(1) 1.23×10^{-7}

(2) 1.17×10^{-4}

(3) 9.35×10^{5}

(4) 9.35×10^{-5}

69. "X" is an oxoanion of the lightest element of group 7 (in the periodic table). The metal is in +6 oxidation state in "X". The color of the potassium salt of X is

(1) green

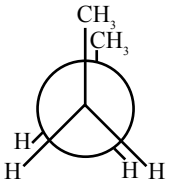
(2) purple

(3) yellow

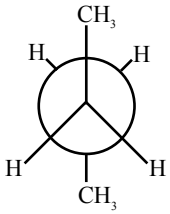
(4) orange



70. Given below are two statements :

Statement I : There are several conformers for n-butane. Out of those conformers,  is

(X)

the least stable and most stable conformer is 

(Y)

Statement II : As the dihedral angle increases, torsional strain decreases from (X) to (Y).

In the light of the above statements, choose the ***correct*** answer from the options given below :

(1) Both Statement I and Statement II are false

(2) Statement I is false but Statement II is true

(3) Statement I is true but Statement II is false

(4) Both Statement I and Statement II are true

SECTION-B

71. 0.25 g of an organic compound "A" containing carbon, hydrogen and oxygen was analysed using the combustion method. There was an increase in mass of $CaCl_2$ tube and potash tube at the end of the experiment. The amount was found to be 0.15 g and 0.1837 g, respectively. The percentage of oxygen in compound A is _____%. (Nearest integer)

(Given : molar mass in g mol^{-1}

H : 1, C : 12, O : 16)

72. The half-life of ^{65}Zn is 245 days. After x days, 75% of original activity remained. The value of x in days is _____. (Nearest integer)

(Given : log 3 = 0.4771 and log 2 = 0.3010)

73. A chromium complex with a formula $CrCl_3.6H_2O$ has a spin only magnetic moment value of 3.87 BM and its solution conductivity corresponds to 1 : 2 electrolyte. 2.75 g of the complex solution was initially passed through a cation exchanger. The solution obtained after the process was reacted with excess of $AgNO_3$. The amount of AgCl formed in the above process is _____ g. (Nearest integer)

[Given : Molar mass in g mol^{-1} Cr : 52; Cl : 35.5, Ag : 108, O : 16, H : 1]

74. Molar conductivity of a weak acid HQ of concentration 0.18 M was found to be 1/30 of the molar conductivity of another weak acid HZ with concentration of 0.02 of M. If $\lambda^0_{Q^-}$ happened to be equal with $\lambda^0_{Z^-}$, then the difference of the pK_a values of the two weak acids (pK_a(HQ)–pK_a(HZ)) is _____ (Nearest integer).

[Given : degree of dissociation (α) << 1 for both weak acids, λ° : limiting molar conductivity of ions]

75. Grignard reagent RMgBr(P) reacts with water and forms a gas (Q). One gram of Q occupies 1.4 dm^3 at STP. (P) on reaction with dry ice in dry ether followed by H_3O^+ forms a compound (Z). 0.1 mole of (Z) will weigh _____ g. (Nearest integer)



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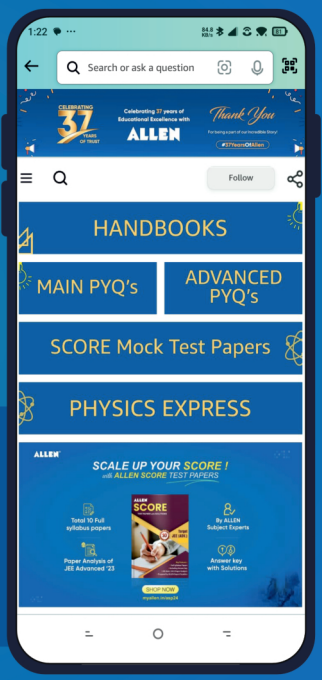
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